HW 6

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1.

Let n_i be the number of students from category i, with $\sum_{i=1}^{5} n_i = 4$

The unordered composition has probability

$$P(n_1, \ldots, n_5) = \frac{4!}{\prod_{i=1}^{5} n_i!} \prod_{i=1}^{5} q_i^{n_i}$$

Laplace smoothing. If the given proportions are empirical frequencies from N students, add-one smoothing gives

$$q_i = \frac{N p_i + 1}{N + 5},$$

$$(p_1, p_2, p_3, p_4, p_5 = 24.22, 20.18.16)$$

The original gives no N, so the smoothed numerical probabilities depend on an unspecified sample size.

(i) All four are fourth-year students: q_4^4

(ii) Two first-year and two second-year students: $\binom{4}{2} q_1^2 q_2^2 = 6 q_1^2 q_2^2$

(iii) The most likely composition is $(n_1, n_2, n_3, n_4, n_5) = (1, 1, 1, 1, 0)$, with probability $24 q_1 q_2 q_3 q_4$.

Indeed, if a category is repeated, there is an unused category. Moving one student from a repeated category i to an unused category j multiplies its probability by $n_i q_j / q_i > 1$, since $n_i \geq 2$ and $q_j / q_i \leq 1.5$. Thus the maximum uses four distinct categories, specifically the four largest.

(iv) The least likely composition is $(0,0,0,0,4)$, with probability q_5^4. Every product is at least q_5^4 and its multinomial coefficient is at least one.

Numerical example, assuming $N = 100$. $q = (25, 23, 21, 19, 17)/105$ gives respectively

0.001072154, 0.016523603, 0.045299643, 0.000687243

For comparison, without smoothing the four probabilities are 0.0010498, 0.01672704, 0.045619920, 0.00065536.

2. By Bayes's rule

$$P(U|R) = \frac{P(R|U)P(U)}{P(R|U)P(U) + P(R|\neg U)P(\neg U)} =$$

$$\boxed{\frac{5}{59}}$$

3. There are 7 mammals and 13 non-mammals.

The counts of each queried attribute value

are:

Class	Birth=yes	fly=yes	water=yes	Legs=yes	Total
M	6	1	2	5	7
N	1	3	3	9	13

Using empirical probabilities and conditional independence

$$S_M = P(M)P(X|M) = \frac{3}{343}$$

$$S_N = P(N)P(X|N) = \frac{81}{43940}$$

Therefore

X is classified as a mammal.

$$P(M|X) = \frac{S_M}{S_M + S_N} = \frac{43940}{53201}$$

4.

From/To	S	C	R	Row total
S	4	1	2	7
C	0	2	2	4
R	2	2	1	5
Total	6	5	5	16

Normalize each row to estimate $A_{ij} = P(W_{t+1} = j \mid W_t = i)$:

$$A = \begin{pmatrix} \frac{4}{7} & \frac{1}{7} & \frac{2}{7} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{2}{5} & \frac{1}{5} \end{pmatrix}$$

5. In state order (S, C, R), is

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

For a window starting on day t,

$$P(W_t = S, W_{t+1} = S, W_{t+2} = S) = P(W_t = S)(.8)$$

solve $\pi A = \pi$, $\sum_i \pi_i = 1$

$$.2\pi_S = .38\pi_C + .75\pi_R$$

$$.4\pi_C = .15\pi_S + .05\pi_R$$

$$\pi = \frac{1}{929}(638, 245, 46).$$

$$\therefore \hat{\gamma}_s(.8)^2 = \frac{1020}{23225}$$

6. $P(M, E, A, C) = P(M)\,P(A)\,P(E|M)\,P(C|E, A)$

$P(M = Hi) = P(M = Lo) = \frac{1}{2}$

$P(A = Working) = \frac{25}{40} = \frac{5}{8}$

$P(A = Broken) = \frac{15}{40} = \frac{3}{8}$

M	$P(E = Good \mid M)$	$P(E = Bad \mid M)$
Hi	$\frac{1}{2}$	$\frac{1}{2}$
Lo	$\frac{3}{4}$	$\frac{1}{4}$

E	A	$P(C=Hi \mid E, A)$	$P(C=Lo \mid E, A)$
Good	Working	$\frac{3}{4}$	$\frac{1}{4}$
Good	Broken	$\frac{2}{3}$	$\frac{1}{3}$
Bad	Working	$\frac{2}{9}$	$\frac{7}{9}$
Bad	Broken	0	1

$$\therefore P(E=Bad, A=Broken) = \frac{3}{8}\left(\frac{1}{2}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{4}\right) = \frac{9}{64}$$

7.

$$P(B,F,G,S) = P(B)P(F)P(G \mid B,F)P(S \mid B,F)$$

$P(B=bad)=.1$, $P(B=good)=.9$

$P(F=empty)=.2$, $P(F=not\ empty)=.8$

(a) $P(S=yes \mid B=good, F=empty) = 1 - 0.8 = .2$

$P(B=good, F=empty, G=empty, S=yes)$

$= 0.9 \times 0.2 \times 0.8 \times 0.2 = 0.0288$

(b) $P(B=bad, F=empty, G=not\ empty, S=no)$

$= 0.1 \times 0.2 \times 0.1 \times 1 = 0.002$

(c) $P(S=yes \mid B=bad) = \sum_f P(S=yes \mid Bad, f) P(f)$

$= (-0.9) \times 0.8 + 0 \times 0.2 = 0.08$